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ABOUT THIS CODELAB

Last updated Mar 4, 2026

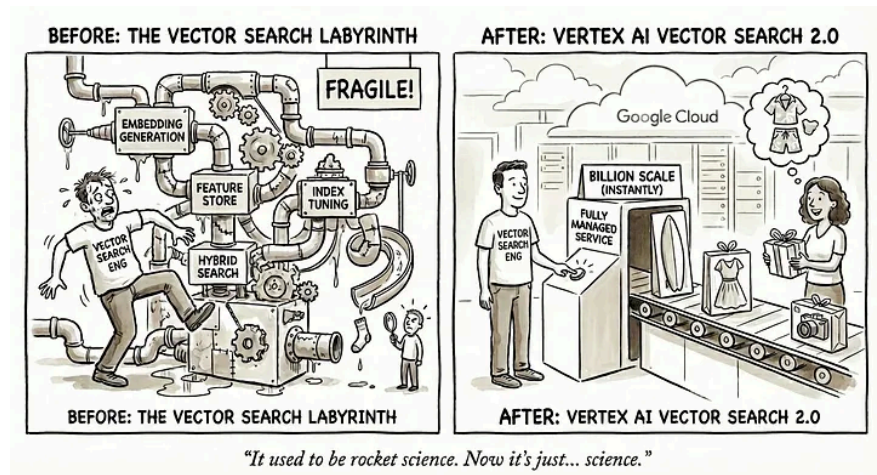
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1. Introduction (#0)

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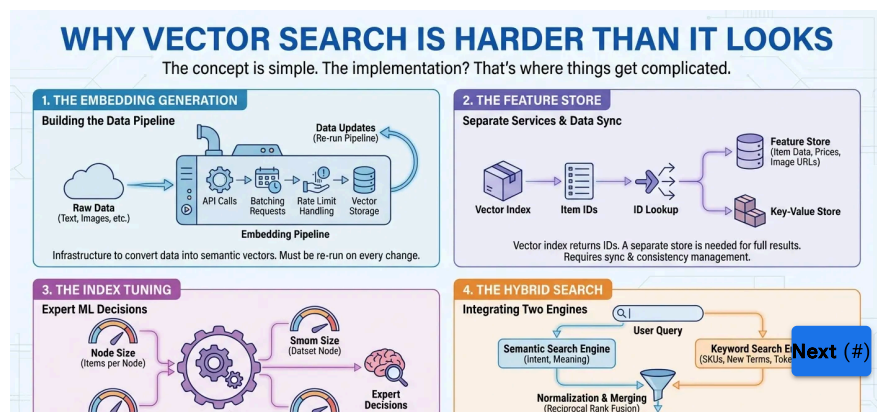
Vector search (https://en.wikipedia.org/wiki/Vector_database), or Vector database, has become a foundational technology for modern AI systems. By representing data as high-dimensional embeddings that capture semantic meaning, it powers everything from semantic search that understands user intent, to recommendation engines that surface relevant content, to Retrieval-Augmented Generation (RAG) and AI Agents that ground LLM responses in real, up-to-date information. Major tech companies including Google rely on this technology at massive scale to process billions of searches, recommendations and groundings daily.

Yet building production-ready vector search remains challenging. Google recently released Vertex AI Vector Search 2.0 (<https://cloud.google.com/vertex-ai/docs/vector-search-2/overview>) to change that — a fully managed service designed to eliminate the design and operational complexity that slows teams down.



Why Vector Search Is Harder Than It Looks

The concept is simple. The implementation? That's where things get complicated.



Report a mistake (#)